No of additional sheets

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

LOURDES HILLS, KUTTICHAL P. O., THIRUVANANTHAPURAMM - 695 574

17/30

ANSWER BOOK

Name Vipin Kumar. N Class No. 33 Branch MCA

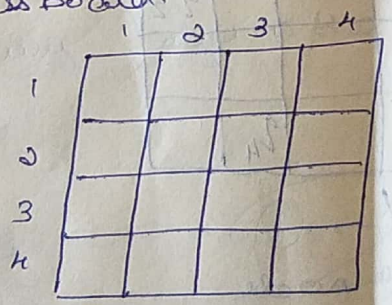
Subject Design & analysis of Algorithm Semester : S3

Mark Tabulation

Mark Divisions		Q. 1	Q. 2	Q. 3	Q. 4	Q. 5	Q. 6	Q. 7	Q. 8	Q. 9	Q. 10	Q. 11	Q. 12	Q. 13	Q. 14	Q. 15
Total																
Grand Total																

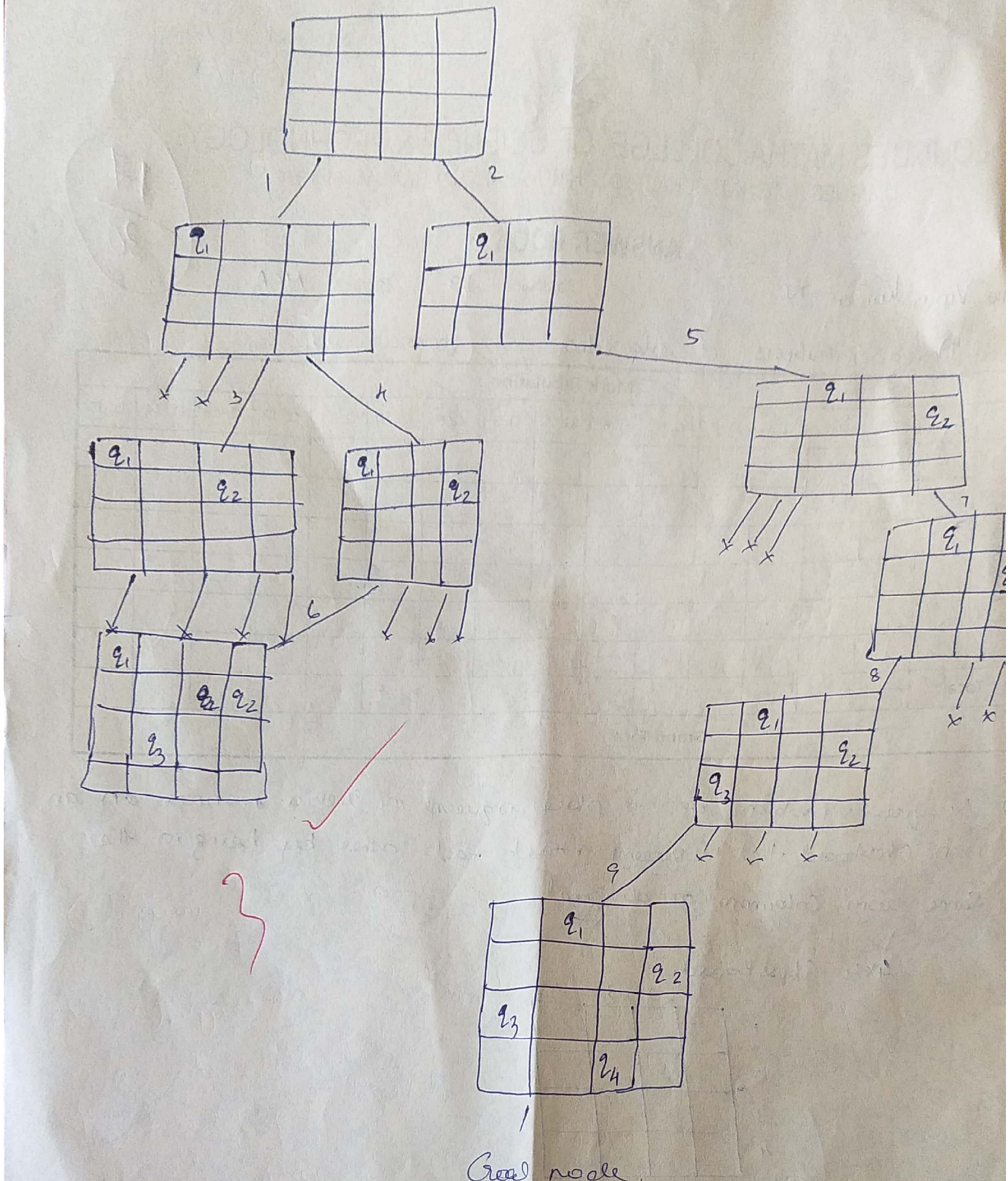
2) N-queens problem is to place n queens in such a manner on an nxn chessboard that no queens attack each other by being in the same row, column or diagonal

4x4 Chessboard.



Here in 4 queens problem we have 4 queens and we need to place it accordingly which satisfy the condition.

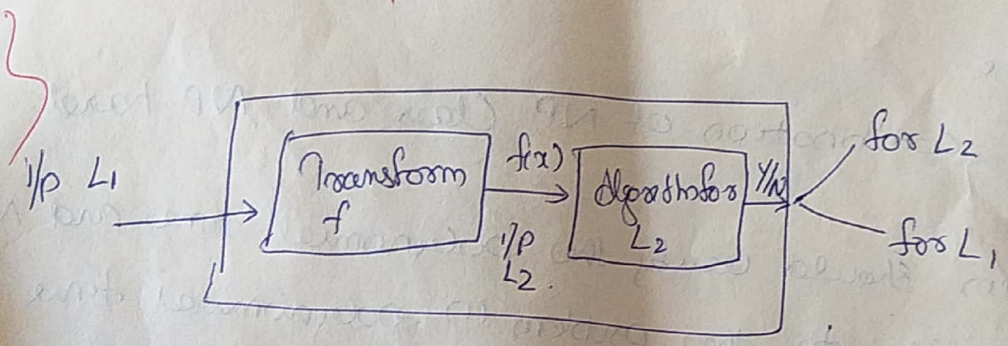
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Goal nodes (2, 5, 7, 8, 9)

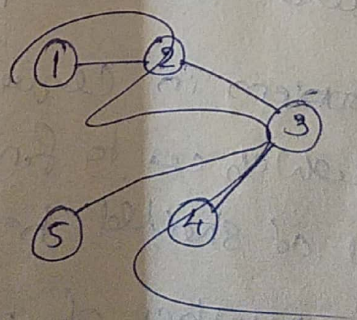
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Polynomial time reduction is a computational complexity theory and its a method for solving the problem using the other. the if the both of the time required to transform the first problem to the second. and the number of times the subroutine is called polynomial. for eg we have two problems A and B. and A is P Class problem if A is reducible to B. and B is a P class problem. So the reduction of A requires polynomial amount of time.



5) Vertex Cover Problem. for a given graph G(V,E) a. two integer value k. the problem is to find whether there a subset V' of the of vertices of size atmost k, such that every edge in the graph is connected to some vertex in V.

for eg:

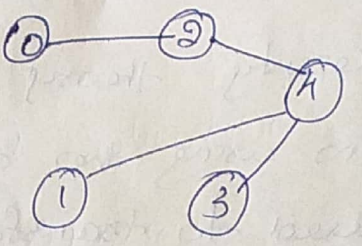


G(V,E)

V = {0,1,2,3,4,5}

E = { <0,3>, <1,3>, <2,3> }

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$G(V, E)$

$V = \{0, 1, 2, 3\}$

$E = \{\langle 0,3\rangle, \langle 1,3\rangle, \langle 2,3\rangle\}$

$V' = \{3\}$

Part B

9) NP- Complete.

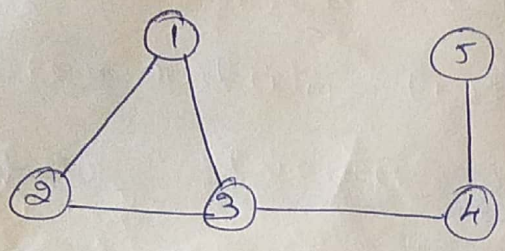
- It is a combination of NP Class and NP-hard problems
- The problem should verify the polynomial time and NP Class reduced to the problem in polynomial time.

Clique Problem

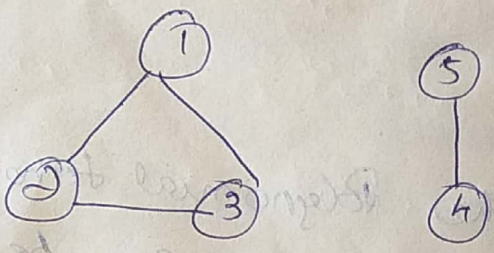
A Clique is a subgraph of a graph, where all vertices is a sub graph that are connected to each other. That means the subgraph should be a complete graph.

- The optimization problem in Clique problem is the Maximal Clique problem which is to find the max sized clique of the graph and it should be a complete graph which contain maximum number of vertices.

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G



A B

These are the derived subgraphs which are complete graph A and B from graph G.

The clique size of the graphs can be identified by counting no. of vertices on those graphs.

for A - its 3

B - its 2.

for a problem to be NP Complete the problem should be NP and NP hard.

Part A

(6) Classes are made according to their time and space complexity of a problem to find its solution and verify it.

Time complexity refers to the how much time it

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takes to solve the problem. the memory usage is
as space complexity.

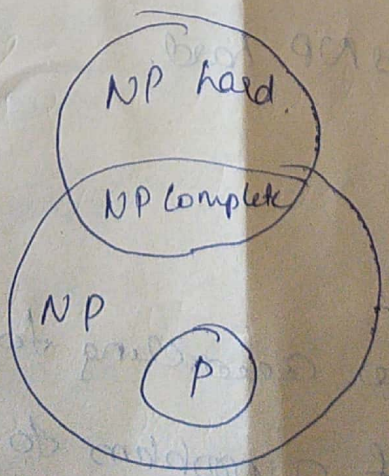
There are 4 Classes

- P Class
- NP Class
- NP hard
- NP Complete.

The P class stands for the Polynomial time. It is the collection of decision problems that can be solved by a deterministic machine in polynomial time. Solution is easy to find.

NP Class stands for Non deterministic Polynomial time its the collection of decision problems that can be solved by a non-deterministic machine in polynomial time.

For this the problem is very hard and its easy to verify solution.



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Part B

⑥ Travelling Salesman Problem.

Given a set of Cities and distance b/w every pair of cities the problem is to find the shortest route so that all City is covered exactly once and return to starting point.

Problem.

Consider a directed graph $G(V, E)$ and each edge have cost C_{ij}. and $|V| = n$ and $n > 1$. The Cost of travel/ tour is simple cycle that include all V. and Cost of tour is cost of all edge cost. The problem is to find the minimum cost route.

Part - A

① Backtracking in which it checks all the possible solution and checks the tree for solution and when it reaches an end and the decision is wrong it backtracks the tree and checks another route till solution is found.

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Branch bound is used for optimization. and in which the optimal solution is found by ignoring the path currently going by when it has much better soln than this path.

it traverses tree in Bfs or Dfs
But Backtracking traverses Dfs way
Backtracking have feasibility fun whereas bound function Branch bound.

Backtracking gives decision problem whereas branch bound solves optimization problems and it is less efficient than backtracking.

(3) Decision tree is a full binary tree that represents the comparisons b/w elements that are performed by a particular sorting algorithm operating on given i/p size.

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